

Kinetics

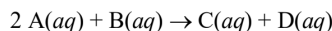
1. Nitrogen dioxide decomposes to nitric oxide and oxygen according to the following reaction:



If $\Delta[\text{NO}_2]/\Delta t$ is -0.0024 M s^{-1} , what is $\Delta[\text{O}_2]/\Delta t$?

- (A) -0.0048 M s^{-1} (B) -0.0012 M s^{-1}
(C) 0.0048 M s^{-1} (D) 0.0012 M s^{-1}

2. For the reaction below, initial rate data appear in the table. What is the rate law for the reaction?



Experiment	$[\text{A}]_0, \text{M}$	$[\text{B}]_0, \text{M}$	Initial rate, M s^{-1}
1	0.10	0.10	0.086
2	0.20	0.10	0.342
3	0.20	0.20	0.344

- (A) $\text{Rate} = k[\text{A}][\text{B}]$ (B) $\text{Rate} = k[\text{A}]^2$
(C) $\text{Rate} = k[\text{A}]^2[\text{B}]$ (D) $\text{Rate} = k[\text{A}][\text{B}]^2$

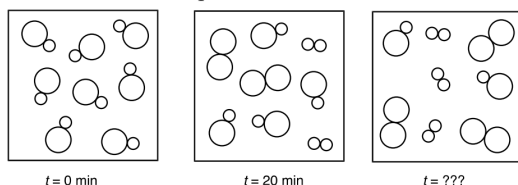
3. Two reactions have the same rate constant at 300.0 K . Reaction 1 has an activation energy of 48.2 kJ/mol , while reaction 2 has an activation energy of 20.0 kJ/mol . What is k_1/k_2 at 320.0 K ?

- (A) 0.493 (B) 0.844
(C) 1.18 (D) 2.03

4. The reaction between A and B has the rate law $\text{Rate} = k[\text{A}][\text{B}]$. A reaction is conducted with $[\text{A}]_0 = 0.0245 \text{ M}$ and $[\text{B}]_0 = 3.24 \times 10^{-5} \text{ M}$, and the natural logarithm of the concentration of $[\text{B}]$ is plotted as a function of time in seconds. The plot is linear, with a slope of -3.96×10^{-4} . What is the value of k ?

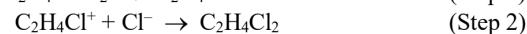
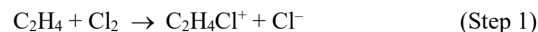
- (A) $3.96 \times 10^{-4} \text{ M}^{-1} \text{ s}^{-1}$ (B) $1.62 \times 10^{-2} \text{ M}^{-1} \text{ s}^{-1}$
(C) $12.2 \text{ M}^{-1} \text{ s}^{-1}$ (D) $499 \text{ M}^{-1} \text{ s}^{-1}$

5. In the depiction of an irreversible chemical reaction shown below, the large circles represent atoms of element A and the small circles represent atoms of element B. At $t = 0$, only compound AB is present. The rate law of the reaction is $\text{Rate} = k[\text{AB}]^2$. What is the value of t in the third panel?



- (A) 30 min (B) 40 min
(C) 50 min (D) 60 min

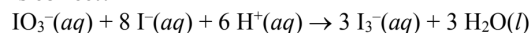
6. The chlorination of ethene is proposed to occur by the following mechanism:



Which step could possibly be rate-determining, given a rate law of $\text{Rate} = k[\text{C}_2\text{H}_4][\text{Cl}_2]$?

- (A) Step 1 only
(B) Step 2 only
(C) Either Step 1 or Step 2
(D) Neither Step 1 nor Step 2

7. When iodate ion reacts with iodide ion in acidic solution, which statement about how concentrations are changing is correct?



- (A) $8 \frac{\Delta[\text{I}^-]}{\Delta t} = 3 \frac{\Delta[\text{I}_3^-]}{\Delta t}$ (B) $8 \frac{\Delta[\text{I}^-]}{\Delta t} = -3 \frac{\Delta[\text{I}_3^-]}{\Delta t}$
(C) $3 \frac{\Delta[\text{I}^-]}{\Delta t} = 8 \frac{\Delta[\text{I}_3^-]}{\Delta t}$ (D) $3 \frac{\Delta[\text{I}^-]}{\Delta t} = -8 \frac{\Delta[\text{I}_3^-]}{\Delta t}$

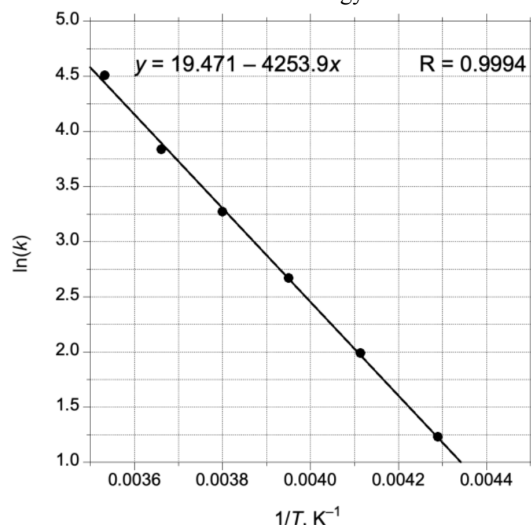
8. For the irreversible reaction between $\text{CO}(\text{g})$ and $\text{Cl}_2(\text{g})$ to form $\text{COCl}_2(\text{g})$, the rate law is $\text{Rate} = k[\text{CO}][\text{Cl}_2]$. In an experiment where equimolar quantities of $\text{CO}(\text{g})$ and $\text{Cl}_2(\text{g})$ are reacted, which of the following would change linearly with reaction time?

- (A) $\ln([\text{CO}])$ (B) $\ln([\text{COCl}_2])$
(C) $1/[\text{CO}]$ (D) $1/[\text{COCl}_2]$

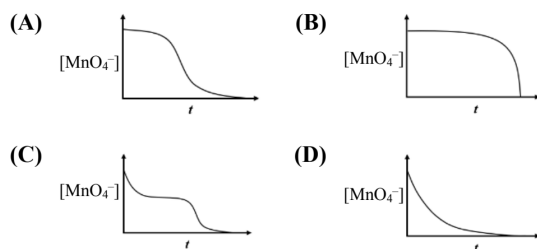
9. A sample containing radium-226 was measured in 1925 to have an activity of 3.74×10^6 disintegrations per second. The same sample, re-measured in 2025, has an activity of 3.58×10^6 disintegrations per second. What is the half-life of ^{226}Ra ?

- (A) 620 years (B) 1200 years
(C) 1600 years (D) 2300 years

10. The rate constant for the addition of pyridine to an iridium compound is measured at a variety of temperatures, and the variation of $\ln(k)$ with $1/T$ is shown below. What is the activation energy of this reaction?



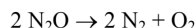
- (A) 1.82 kJ mol^{-1} (B) 4.25 kJ mol^{-1}
 (C) 35.4 kJ mol^{-1} (D) 82.8 kJ mol^{-1}
11. Cyclopropane is proposed to isomerize to propene through an intermediate diradical as shown below. The overall reaction is observed to follow the rate law $\text{Rate} = k[\text{cyclo-C}_3\text{H}_6]$. If k_2 is much smaller than k_{-1} , what is the value of k in terms of k_1 , k_{-1} , and k_2 ?
- $$\text{cyclo-C}_3\text{H}_6 \xrightleftharpoons[k_{-1}]{k_1} \cdot\text{CH}_2\text{CH}_2\text{CH}_2\cdot$$
- $$\cdot\text{CH}_2\text{CH}_2\text{CH}_2\cdot \xrightarrow{k_2} \text{CH}_3\text{CH}=\text{CH}_2$$
- (A) k_1 (B) k_1/k_{-1}
 (C) k_1k_2 (D) k_1k_2/k_{-1}
12. The reaction of $\text{MnO}_4^-(aq)$ and $\text{H}_2\text{C}_2\text{O}_4(aq)$ in acidic solution is autocatalytic because one of the products, $\text{Mn}^{2+}(aq)$, is a reducing agent. Starting from a mixture of reactants, which of the following graphs best shows how the concentration of $\text{MnO}_4^-(aq)$ changes over time?



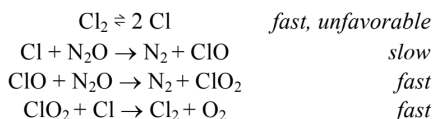
13. Ethanol ($\text{C}_2\text{H}_5\text{OH}$) is oxidized to acetaldehyde (CH_3CHO) by dichromate ions in acidic solution according to the following equation:
- $$3 \text{C}_2\text{H}_5\text{OH}(aq) + \text{Cr}_2\text{O}_7^{2-}(aq) + 8 \text{H}^+(aq) \rightarrow 3 \text{CH}_3\text{CHO}(aq) + 2 \text{Cr}^{3+}(aq) + 7 \text{H}_2\text{O}(l)$$
- If the concentration of $\text{Cr}^{3+}(aq)$ is increasing at a rate of 0.18 M s^{-1} , at what rate is the concentration of $\text{CH}_3\text{CHO}(aq)$ changing?
- (A) Increasing at 0.12 M s^{-1}
 (B) Increasing at 0.27 M s^{-1}
 (C) Decreasing at 0.12 M s^{-1}
 (D) Decreasing at 0.27 M s^{-1}
14. A reaction in the gas phase doubles in rate on increasing the temperature from 10°C to 20°C . Which is the best explanation for the increase in reaction rate?
- (A) The pressure doubles.
 (B) The number of particles above a minimum activation energy doubles.
 (C) The average kinetic energy of the molecules doubles.
 (D) The number of collisions per unit time doubles.
15. An irreversible reaction occurs as follows:
- $$2 \text{A}(g) + \text{B}(g) \rightarrow \text{C}(g) + \text{D}(g)$$
- What is the rate law for this reaction?
- (A) $\text{Rate} = k[\text{A}][\text{B}]$
 (B) $\text{Rate} = k[\text{A}]^2[\text{B}]$
 (C) $\text{Rate} = k[2\text{A}][\text{B}]$
 (D) It cannot be determined from the information given.
16. The half-life of ^{90}Sr is 28 years. How long is required for the level of radioactivity to drop to 1.5% of its initial value?
- (A) 0.42 years (B) 40 years
 (C) 55 years (D) 170 years
17. An uncatalyzed reaction has an activation energy $E_a = 50 \text{ kJ mol}^{-1}$ and an enthalpy change $\Delta H^\circ = -10 \text{ kJ mol}^{-1}$. In the presence of a catalyst, what might these values become?
- (A) $E_a = 30 \text{ kJ mol}^{-1}$, $\Delta H^\circ = -10 \text{ kJ mol}^{-1}$
 (B) $E_a = 50 \text{ kJ mol}^{-1}$, $\Delta H^\circ = -15 \text{ kJ mol}^{-1}$
 (C) $E_a = 60 \text{ kJ mol}^{-1}$, $\Delta H^\circ = -20 \text{ kJ mol}^{-1}$
 (D) $E_a = -10 \text{ kJ mol}^{-1}$, $\Delta H^\circ = 50 \text{ kJ mol}^{-1}$

Kinetics

18. Chlorine gas catalyzes the gas-phase decomposition of nitrous oxide according to the equation



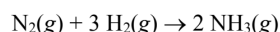
The following mechanism is proposed for the catalyzed reaction:



What rate law is predicted by this mechanism?

- (A) Rate = $k[\text{N}_2\text{O}]$ (B) Rate = $k[\text{N}_2\text{O}]^2$
(C) Rate = $k[\text{N}_2\text{O}][\text{Cl}_2]$ (D) Rate = $k[\text{N}_2\text{O}][\text{Cl}_2]^{1/2}$

19. Hydrogen and nitrogen react according to the following equation:



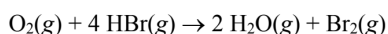
If the concentration of $\text{H}_2(\text{g})$ is changing at a rate of $-12 \text{ mol L}^{-1} \text{ s}^{-1}$, at what rate is the concentration of $\text{NH}_3(\text{g})$ changing?

- (A) $-6.0 \text{ mol L}^{-1} \text{ s}^{-1}$ (B) $-4.0 \text{ mol L}^{-1} \text{ s}^{-1}$
(C) $+8.0 \text{ mol L}^{-1} \text{ s}^{-1}$ (D) $+12 \text{ mol L}^{-1} \text{ s}^{-1}$

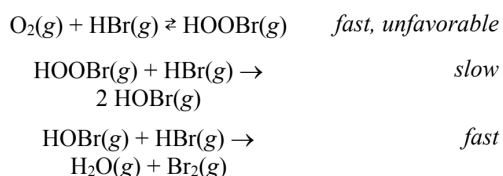
20. An irreversible reaction $\text{R} \rightarrow \text{P}$ has the rate law Rate = $k[\text{R}]^2$. Which graph will be a straight line?

- (A) $\ln([\text{R}])$ as a function of t
(B) $[\text{R}]$ as a function of $\ln(t)$
(C) $1/[\text{R}]$ as a function of t
(D) $[\text{R}]$ as a function of $1/t$

21. For the reaction



the following mechanism is proposed:



What rate law is predicted by this mechanism?

- (A) Rate = $k[\text{HBr}]$ (B) Rate = $k[\text{O}_2][\text{HBr}]$
(C) Rate = $k[\text{O}_2][\text{HBr}]^2$ (D) Rate = $k[\text{O}_2][\text{HBr}]^4$

22. The radioisotope $^{99\text{m}}\text{Tc}$ undergoes gamma decay with a rate constant $k = 0.116 \text{ h}^{-1}$. How much time is required for its activity to decrease to 10.0% of its original activity?

- (A) 19.8 h (B) 7.76 h (C) 5.98 h (D) 1.16 h

23. A cotton ball burns more brightly in a jar of 80% O_2 than in the air, which is 20% O_2 . Which is the best explanation for this observation?

- (A) The cotton collides more frequently with the O_2 inside the jar.
(B) The excited state of O_2 is formed more efficiently inside the jar.
(C) The temperature is higher inside the jar.
(D) The jar acts as a catalyst for the reaction.

24. A reaction has an activation energy of 75.0 kJ mol^{-1} . At what temperature will the rate constant be twice as large as it is at 298 K?

- (A) 305 K (B) 307 K (C) 311 K (D) 319 K

25. *Tert*-butyl chloride reacts with hydroxide ion in a process that is first order in both *tert*-butyl chloride and hydroxide. If both reactants are doubled in concentration, how does the reaction rate change?

- (A) The rate does not change.
(B) The rate doubles.
(C) The rate quadruples.
(D) The effect on the rate cannot be determined from the information given.

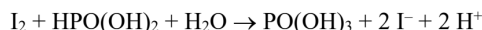
26. Under certain conditions, the rate of decomposition of hydrogen peroxide is first order in H_2O_2 . It is observed that the concentration of H_2O_2 decreases by 1.00% in 300.0 s. What is the rate constant for the reaction?

- (A) $3.35 \times 10^{-5} \text{ s}^{-1}$ (B) $2.31 \times 10^{-3} \text{ s}^{-1}$
(C) $3.33 \times 10^{-3} \text{ s}^{-1}$ (D) 300 s^{-1}

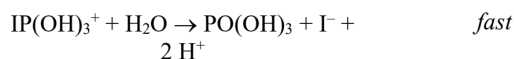
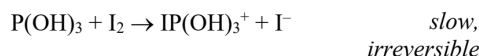
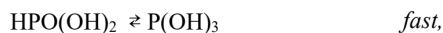
27. The pain reliever naproxen has a half-life of 12 h in the human body. A patient takes a 400 mg tablet of naproxen at 8 AM and a second 400 mg tablet at 8 PM (12 h after the first dose). How much naproxen remains in the patient's body at 8 AM the following day (12 h after the second dose)?

- (A) 50 mg (B) 100 mg
(C) 200 mg (D) 300 mg

28. For the aqueous reaction



the following mechanism is proposed:



Which observation is consistent with this mechanism?

- (A) The reaction is first order in both $\text{HPO}(\text{OH})_2$ and I_2 .
- (B) The reaction rate is independent of the I_2 concentration.
- (C) The reaction goes faster as the concentration of H^+ is increased.
- (D) The reaction goes slower as the concentration of I^- is increased.

29. In which ways may a catalyst increase the rate of a reaction?

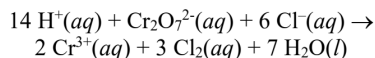
- I. It may alter the rate law of the reaction.
- II. It may make the overall reaction more exothermic.

- (A) I only
- (B) II only
- (C) Both I and II
- (D) Neither I nor II

30. The rate constant of a chemical reaction increases by 63% when the temperature is raised from 40 °C to 45 °C. What is the activation energy of this reaction?

- (A) 1.5 kJ mol⁻¹
- (B) 9.7 kJ mol⁻¹
- (C) 76.5 kJ mol⁻¹
- (D) 80.9 kJ mol⁻¹

31. Dichromate ion is reduced by chloride ion according to the equation below. If $[\text{Cl}^-]$ is decreasing at a rate of 0.37 mol L⁻¹ min⁻¹, how is the concentration of $\text{Cr}^{3+}(\text{aq})$ changing?



- (A) Increasing by 0.12 mol L⁻¹ min⁻¹
- (B) Increasing by 1.11 mol L⁻¹ min⁻¹
- (C) Decreasing by 0.12 mol L⁻¹ min⁻¹
- (D) Decreasing by 1.11 mol L⁻¹ min⁻¹

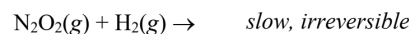
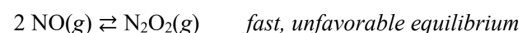
32. The isotope cobalt-57 has a half-life of 272 d. How long will it take a sample of ⁵⁷Co to decay to 38% of its original activity?

- (A) 103 d
- (B) 337 d
- (C) 188 d
- (D) 380 d

33. Which statement best describes the variation of the rate constant of a reaction with temperature?

- (A) The rate constant does not change with temperature because it is an unvarying characteristic of the specific reaction.
- (B) The rate constant typically decreases with increasing temperature because fewer molecules are able to adopt the required orientation at higher temperature.
- (C) The rate constant typically increases with increasing temperature because increasing the temperature increases the fraction of collisions that result in reaction.
- (D) The rate constant typically increases with increasing temperature because most reactions become more favorable as the temperature increases.

34. A mechanism for the reaction of nitric oxide with hydrogen to form water and nitrogen gas is proposed below. What rate law is predicted by this mechanism?



- (A) Rate = $k[\text{NO}]^2$
- (B) Rate = $k[\text{NO}]^2[\text{H}_2]$
- (C) Rate = $k[\text{NO}]^2[\text{H}_2]^2$
- (D) Rate = $k \frac{[\text{NO}]^2[\text{H}_2]^2}{[\text{N}_2\text{O}][\text{H}_2\text{O}]}$

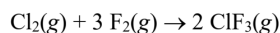
35. An irreversible reaction $\text{A} + \text{B} \rightarrow \text{C}$ is carried out under various conditions, but always with $[\text{B}]_0 \gg [\text{A}]_0$. Plots of $1/[\text{A}]$ as a function of time for each reaction are linear, and the slopes of these plots do not depend on the initial concentration of B. What is the rate law for the reaction under these conditions?

- (A) Rate = $k[\text{A}]$
- (B) Rate = $k[\text{A}][\text{B}]$
- (C) Rate = $k[\text{A}]^2$
- (D) Rate = $k[\text{A}]^2[\text{B}]$

36. The reaction of $\text{H}_2(\text{g})$ with $\text{C}_2\text{H}_4(\text{g})$ to form $\text{C}_2\text{H}_6(\text{g})$ is catalyzed by metallic platinum. Which statement most accurately describes the impact of changing the catalyst in this reaction?

- (A) The rate of the reaction would not change if a greater mass of catalyst were used because the platinum is a solid so its concentration is a fixed number.
- (B) The rate of the reaction would increase if more finely divided platinum were used because the reaction depends on the surface area of the catalyst.
- (C) The ΔH° for the reaction would become more negative if a greater mass of catalyst were used because the catalyst adsorbs the reactants exothermically.
- (D) The ΔS° for the reaction would become more negative if more finely divided platinum were used because the catalyst absorbs gas-phase reactants to form ordered surface species.

37. In the reaction



the rate of disappearance of $\text{F}_2(\text{g})$ is 1.0 M s^{-1} . What is the rate of appearance of $\text{ClF}_3(\text{g})$?

- (A) 0.33 M s^{-1}
- (B) 0.67 M s^{-1}
- (C) 1.0 M s^{-1}
- (D) 1.5 M s^{-1}

38. When the rate of the reversible reaction $\text{A} + \text{B} \rightleftharpoons \text{C}$ is studied under a certain set of conditions, it is found that the rate of the forward reaction is $k_f[\text{A}]$. What can be concluded about the rate law for the reverse reaction under these conditions?

- (A) $\text{Rate} = k_r[\text{C}]$
- (B) $\text{Rate} = k_r \frac{[\text{C}]}{[\text{B}]}$
- (C) The rate law of the reverse reaction cannot be determined from the information given.
- (D) An error must have been made, since if the reaction is reversible, the forward rate law must be $\text{Rate} = k_f[\text{A}][\text{B}]$.

39. A substance A decomposes irreversibly to form B. A plot of $\ln([\text{A}])$ as a function of time from the beginning of the reaction until A is 97% consumed is a straight line with a negative slope. What is the reaction order in A?

- (A) Zero order
- (B) First order
- (C) Second order
- (D) Third order

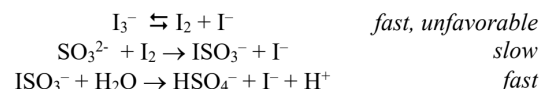
40. Which statement about catalysis is correct?

- (A) If a catalyst increases the forward rate of a reaction by a factor of two, it must increase the rate of the reverse reaction by a factor of two.
- (B) If a catalyst increases the rate of formation of a product by a factor of two, it must increase the rate of formation of the mirror image of the product by a factor of two.
- (C) A catalyst must be in the same phase as the reactants and products of the reaction.
- (D) A catalyzed reaction must proceed by the same mechanism as the uncatalyzed reaction, but with a lower activation energy.

41. The isotope ^{226}Ra has a half-life for radioactive decay of 1600 y. How long will it take the amount of ^{226}Ra in a sample of $^{226}\text{RaCl}_2$ to decrease by 25%?

- (A) 660 y
- (B) 800 y
- (C) 1200 y
- (D) 1600 y

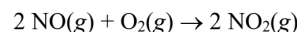
42. The oxidation of sulfite ion by triiodide ion is proposed to take place by the following mechanism:



What rate law is predicted by this mechanism?

- (A) $\text{Rate} = k[\text{I}_3^-]$
- (B) $\text{Rate} = k[\text{I}_3^-][\text{SO}_3^{2-}]$
- (C) $\text{Rate} = \frac{k[\text{I}_3^-][\text{SO}_3^{2-}]}{[\text{I}^-]}$
- (D) $\text{Rate} = \frac{k[\text{I}_3^-][\text{SO}_3^{2-}]}{[\text{H}^+]}$

43. In a study of the reaction below, the concentration of $\text{O}_2(\text{g})$ is found to be decreasing by 0.042 M min^{-1} . At what rate is the concentration of nitrogen dioxide gas changing?



- (A) Increasing by 0.021 M min^{-1}
- (B) Increasing by 0.042 M min^{-1}
- (C) Increasing by 0.084 M min^{-1}
- (D) It cannot be determined without knowing the rate law for the reaction.

44. The rate constant for an elementary chemical reaction can be affected by which of the following?

- I. Reactant concentrations
- II. Product concentrations

- (A) I only
- (B) II only
- (C) Both I and II
- (D) Neither I nor II

Kinetics

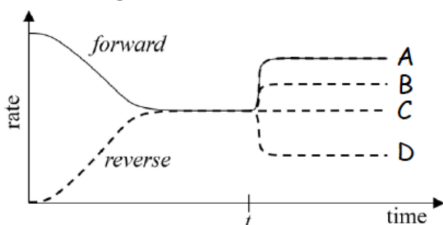
45. Sulfuryl chloride (SO_2Cl_2) decomposes via first-order kinetics. The half-life is 4.1 minutes at a certain temperature. How long does it take for 30% of the SO_2Cl_2 in a sample to decompose at this temperature?

(A) 0.6 min (B) 2.1 min
(C) 2.5 min (D) 7.1 min

46. The rate constant of a chemical reaction increases 26% when the temperature is raised from 50°C to 55°C . What is the activation energy of this reaction?

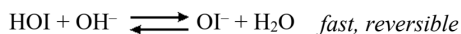
(A) 1.1 kJ mol^{-1} (B) 18 kJ mol^{-1}
(C) 41 kJ mol^{-1} (D) 220 kJ mol^{-1}

47. The graph below shows the forward and reverse rates for a reaction as a function of time. At time t , a catalyst is added to the system, and the forward reaction rate is observed to change as indicated by the solid curve. Which dashed curve best indicates how the reverse reaction rate changes?



(A) A (B) B (C) C (D) D

48. The following mechanism is proposed for the oxidation of I^- by OCl^- in aqueous solution:



What reaction orders for OCl^- , I^- , and OH^- are consistent with this mechanism?

	Order in OCl^-	Order in I^-	Order in OH^-
(A)	1	1	1
(B)	1	1	0
(C)	1	0	0
(D)	1	1	-1

49. The following data were collected for the reaction of A and B:

$[\text{A}]_0 \text{ (M)}$	$[\text{B}]_0 \text{ (M)}$	Initial rate (M/s)
0.20	0.20	600
0.60	0.40	7200
0.60	0.80	28800

What is the order of the reaction with respect to A and B?

(A) 0th order in A, 2nd order in B
(B) 1st order in A, 1st order in B
(C) 2nd order in A, 1st order in B
(D) 1st order in A, 2nd order in B

50. A sample of 0.900 mol N_2O is placed in a sealed container, where it decomposes irreversibly to N_2 and O_2 in a first-order reaction. After 42.0 min, 0.640 mol N_2O remains. How long will it take for the reaction to be 90.0% complete?

(A) 13.0 min (B) 85.4 min
(C) 131 min (D) 284 min

51. The rate of a reaction is 2.3 times faster at 60°C than it is at 50°C . By what factor will the rate increase on going from 60°C to 70°C ?

(A) By less than a factor of 2.3
(B) By a factor of 2.3
(C) By more than a factor of 2.3
(D) The rate increase cannot be determined from the information given.

52. How may an enzyme inhibitor decrease the rate of a reaction catalyzed by the enzyme?
- The inhibitor may bind to the enzyme and block the active site.
 - The inhibitor may shift the equilibrium of the reaction toward the starting materials.

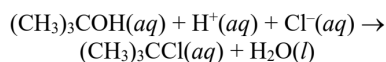
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

53. An irreversible reaction, $\text{A} + \text{B} \rightarrow \text{products}$, is studied under conditions where $[\text{B}] \gg [\text{A}]$. When $[\text{B}] = 0.10 \text{ M}$, a plot of $1/[\text{A}]$ vs. time is linear. When $[\text{B}] = 0.30 \text{ M}$, the plot of $1/[\text{A}]$ vs. time is indistinguishable from the first plot within experimental error. What is the rate law for this reaction?

(A) $\text{Rate} = k[\text{A}][\text{B}]$ (B) $\text{Rate} = k[\text{A}]^2$
(C) $\text{Rate} = k[\text{A}]^2[\text{B}]$ (D) $\text{Rate} = k[\text{A}][\text{B}]^2$

Kinetics

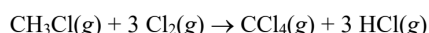
54. *Tert*-butyl alcohol reacts with aqueous hydrochloric acid to give *tert*-butyl chloride.



For this reaction, $\text{Rate} = k[(\text{CH}_3)_3\text{COH}][\text{H}^+]$. With which mechanism is this rate law consistent?

- | | |
|--|-------------|
| (A) $(\text{CH}_3)_3\text{COH}(aq) + \text{H}^+(aq) \rightleftharpoons (\text{CH}_3)_3\text{C}(\text{OH}_2)^+(aq)$ | <i>fast</i> |
| $(\text{CH}_3)_3\text{C}(\text{OH}_2)^+(aq) + \text{Cl}^-(aq) \rightarrow (\text{CH}_3)_3\text{CCl}(aq) + \text{H}_2\text{O}(l)$ | <i>slow</i> |
| (B) $(\text{CH}_3)_3\text{COH}(aq) + \text{H}^+(aq) \rightleftharpoons (\text{CH}_3)_3\text{C}(\text{OH}_2)^+(aq)$ | <i>fast</i> |
| $(\text{CH}_3)_3\text{C}(\text{OH}_2)^+(aq) \rightarrow (\text{CH}_3)_3\text{C}^+(aq) + \text{H}_2\text{O}(l)$ | <i>slow</i> |
| $(\text{CH}_3)_3\text{C}^+(aq) + \text{Cl}^-(aq) \rightarrow (\text{CH}_3)_3\text{CCl}(aq)$ | <i>fast</i> |
| (C) $(\text{CH}_3)_3\text{COH}(aq) + \text{Cl}^-(aq) \rightarrow (\text{CH}_3)_3\text{CCl}(aq) + \text{OH}^-(aq)$ | <i>slow</i> |
| $\text{OH}^-(aq) + \text{H}^+(aq) \rightarrow \text{H}_2\text{O}(l)$ | <i>fast</i> |
| (D) $(\text{CH}_3)_3\text{COH}(aq) \rightarrow (\text{CH}_3)_3\text{C}^+(aq) + \text{OH}^-(aq)$ | <i>slow</i> |
| $(\text{CH}_3)_3\text{C}^+(aq) + \text{Cl}^-(aq) \rightarrow (\text{CH}_3)_3\text{CCl}(aq)$ | <i>fast</i> |
| $\text{OH}^-(aq) + \text{H}^+(aq) \rightarrow \text{H}_2\text{O}(l)$ | <i>fast</i> |

55. Carbon tetrachloride is produced from methyl chloride and chlorine according to the following equation:



If the rate of formation of CCl_4 is measured to be 0.063 M min^{-1} , what is the rate of disappearance of Cl_2 ?

- | | |
|--------------------------------|--------------------------------|
| (A) 0.021 M min^{-1} | (B) 0.063 M min^{-1} |
| (C) 0.13 M min^{-1} | (D) 0.19 M min^{-1} |

56. Iodine-131 decays with a half-life of 8.02 d. In a sample initially containing 5.00 mg of ^{131}I , what mass remains after 6.01 d?

- | | |
|-------------|-------------|
| (A) 1.13 mg | (B) 1.87 mg |
| (C) 2.97 mg | (D) 3.13 mg |

57. For an irreversible reaction $\text{A} \rightarrow \text{products}$, the graph of $1/[\text{A}]$ as a function of time is linear. What is the reaction order in A?

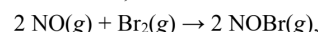
- (A) Zeroth-order
 (B) First-order
 (C) Second-order
 (D) The order in A cannot be determined based on the information given.

58. In comparing two reactions, the reaction with the greater activation energy always has

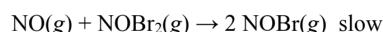
- (A) the slower rate.
 (B) the faster rate.
 (C) the rate that varies less with temperature.
 (D) the rate that varies more with temperature.

Questions 59 and 60 concern the reaction and **mechanism below**.

The formation of NOBr ,



is studied, and the following mechanism is proposed:



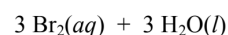
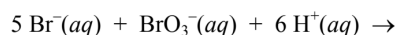
59. What rate law is predicted by this mechanism?

- | | |
|---|---|
| (A) $\text{Rate} = k[\text{NO}][\text{Br}_2]$ | (B) $\text{Rate} = k[\text{NO}]^2[\text{Br}_2]$ |
| (C) $\text{Rate} = k[\text{NO}][\text{Br}_2]^2$ | (D) $\text{Rate} = k[\text{NO}]^2$ |

60. In this reaction, $\text{NOBr}_2(g)$ is best described as

- (A) an intermediate.
 (B) a product.
 (C) a homogeneous catalyst.
 (D) a heterogeneous catalyst.

61. If elemental bromine is being formed according to the equation below at a rate of 0.056 M s^{-1} , at what rate is bromide ion being consumed?

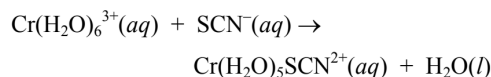


- | | |
|------------------------------|------------------------------|
| (A) 0.019 M s^{-1} | (B) 0.034 M s^{-1} |
| (C) 0.056 M s^{-1} | (D) 0.093 M s^{-1} |

62. Thallium-201, a radioactive isotope used to image the heart, has a half-life of 3.05 d. How long would it take for a sample of thallium-201 to decay to 18% of its original activity?

- | | | | |
|-----------|-----------|-----------|----------|
| (A) 4.4 d | (B) 6.1 d | (C) 7.5 d | (D) 17 d |
|-----------|-----------|-----------|----------|

63. For the reaction:



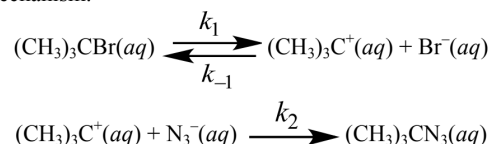
The following data were collected:

$[\text{Cr}(\text{H}_2\text{O})_6^{3+}]$, M	$[\text{SCN}^-]$, M	Rate, M hr^{-1}
0.028	0.040	8.1×10^{-6}
0.028	0.055	1.1×10^{-5}
0.037	0.055	1.5×10^{-5}

What is the rate law for the reaction?

- (A) $\text{Rate} = (9.1 \times 10^{-9} \text{ M}^{-1} \text{ hr}^{-1})[\text{Cr}(\text{H}_2\text{O})_6^{3+}][\text{SCN}^-]$
 (B) $\text{Rate} = (7.2 \times 10^{-3} \text{ M}^{-1} \text{ hr}^{-1})[\text{Cr}(\text{H}_2\text{O})_6^{3+}][\text{SCN}^-]$
 (C) $\text{Rate} = (2.9 \times 10^{-4} \text{ hr}^{-1})[\text{Cr}(\text{H}_2\text{O})_6^{3+}]$
 (D) $\text{Rate} = (3.9 \times 10^{-4} \text{ hr}^{-1})[\text{Cr}(\text{H}_2\text{O})_6^{3+}]$
64. In which of the following ways may a catalyst increase the rate of a reaction?
- I. It may alter the rate law.
 II. It may decrease the overall activation energy.
- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II
65. For an exothermic reaction, which of the following best describes the effect of increasing the temperature on the forward and reverse reactions?
- (A) Both the forward and reverse rates increase, but the forward rate increases more than the reverse rate.
 (B) Both the forward and reverse rates increase, but the reverse rate increases more than the forward rate.
 (C) The forward rate increases while the reverse rate decreases.
 (D) The reverse rate increases while the forward rate decreases.

66. The reaction of *tert*-butyl bromide with azide ion in aqueous solution is proposed to proceed by the following mechanism:

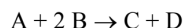


Assuming that $(\text{CH}_3)_3\text{C}^+(\text{aq})$ achieves a steady-state concentration, but making no further assumptions about the relative magnitudes of the three rate constants, what is the rate law for this reaction?

- (A) $\text{Rate} = k_1[(\text{CH}_3)_3\text{CBr}]$
 (B) $\text{Rate} = k_2[(\text{CH}_3)_3\text{CBr}][\text{N}_3^-]$
 (C) $\text{Rate} = \frac{k_1 k_2 [(\text{CH}_3)_3\text{CBr}][\text{N}_3^-]}{k_{-1}[\text{Br}^-]}$
 (D) $\text{Rate} = \frac{k_1 k_2 [(\text{CH}_3)_3\text{CBr}][\text{N}_3^-]}{k_{-1}[\text{Br}^-] + k_2[\text{N}_3^-]}$
67. Which statement about chemical reaction mechanisms is correct?
- (A) The overall rate law can be determined from any step in the mechanism.
 (B) The rate of a reaction is the rate of the fastest elementary step of its mechanism.
 (C) The chemical equation for the sum of all the elementary steps is the chemical equation of the overall reaction.
 (D) Species that are produced and subsequently consumed in the mechanism serve as catalysts for the reaction.
68. At 280 °C, nitrogen dioxide decomposes to nitric oxide and oxygen:
- $$2 \text{NO}_2(\text{g}) \rightarrow 2 \text{NO}(\text{g}) + \text{O}_2(\text{g})$$
- In one experiment, the concentration of NO_2 decreased from 0.0100 M to 0.0050 M over the course of 100. s. What was the average rate of disappearance of $\text{NO}_2(\text{g})$?
- (A) $1.0 \times 10^{-2} \text{ M s}^{-1}$ (B) $1.0 \times 10^{-4} \text{ M s}^{-1}$
 (C) $5.0 \times 10^{-5} \text{ M s}^{-1}$ (D) $2.5 \times 10^{-5} \text{ M s}^{-1}$

Kinetics

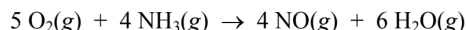
69. What is the rate law for the following reaction?



$[A]_0, M$	$[B]_0, M$	Initial rate, $M s^{-1}$
0.050	0.100	0.085
0.050	0.200	0.170
0.100	0.300	0.510

- (A) Rate = $k[A]$ (B) Rate = $k[B]$
 (C) Rate = $k[A][B]$ (D) Rate = $k[A][B]^2$
70. What are the units of k if the rate law of a reaction is rate = $k[X]^0[Y]^0$?
 (A) $M s^{-1}$ (B) s^{-1}
 (C) $M^{-1} s^{-1}$ (D) k is dimensionless
71. Hydrogen peroxide, $H_2O_2(aq)$, decomposes into water and oxygen. Adding a small amount of $FeCl_3(aq)$ increases the rate of gas evolution in this reaction. What is the best description of the role of $FeCl_3$?
 (A) Transition state
 (B) Reaction intermediate
 (C) Heterogeneous catalyst
 (D) Homogeneous catalyst
72. A substance decomposes in a first-order reaction with a rate constant of $6.70 \times 10^{-4} s^{-1}$. If the initial concentration of the substance is 1.50 M, what is its concentration after 500. s?
 (A) 1.07 M (B) 0.503 M
 (C) 0.335 M (D) 0.128 M

73. For the reaction



if NH_3 is being consumed at a rate of $0.50 M \cdot s^{-1}$, at what rate is H_2O being formed?

- (A) $0.33 M \cdot s^{-1}$ (B) $0.50 M \cdot s^{-1}$
 (C) $0.75 M \cdot s^{-1}$ (D) $3.0 M \cdot s^{-1}$
74. The rate of decomposition of hydrogen peroxide is first order in H_2O_2 . At $[H_2O_2] = 0.150 M$, the decomposition rate was measured to be $4.83 \times 10^{-6} M \cdot s^{-1}$. What is the rate constant for the reaction?
 (A) $2.15 \times 10^{-4} s^{-1}$ (B) $3.22 \times 10^{-5} s^{-1}$
 (C) $4.83 \times 10^{-6} s^{-1}$ (D) $7.25 \times 10^{-7} s^{-1}$

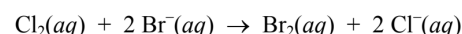
75. In the reaction $A \rightarrow B$, a plot of $1/[A]$ vs. t is found to be linear. What is the reaction order in A?

(A) Zeroth order (B) First order
 (C) Second order (D) Third order

76. The half-life of iodine-131 is 8.02 days. How long will it take for 80.% of the sample to decay?

(A) 2.6 days (B) 13 days
 (C) 19 days (D) 32 days

77. For the reaction



which of the following could be used to monitor the rate?

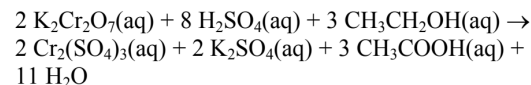
I. pH meter II. Spectrophotometer

(A) I only (B) II only
 (C) Either I or II (D) Neither I nor II

78. For a reversible exothermic reaction, what is the effect of increasing temperature on the equilibrium constant (K_{eq}) and on the forward rate constant (k_f)?

(A) K_{eq} and k_f both increase
 (B) K_{eq} and k_f both decrease
 (C) K_{eq} increases and k_f decreases
 (D) K_{eq} decreases and k_f increases

79. A device used to test alcohol in the bloodstream uses the reaction



If the rate of disappearance of $K_2Cr_2O_7$ is 1.24 M/min what is the rate of disappearance of CH_3CH_2OH ?

(A) 0.415 M/min (B) 0.826 M/min
 (C) 1.86 M/min (D) 3.72 M/min

80. $C(s) + CO_2(g) \rightarrow 2 CO(g)$ $\Delta H = 172 kJ \cdot mol^{-1}$
 In a suitable reaction vessel, pieces of graphite are mixed with carbon dioxide gas at 1.00 atm and 1000 K. Which of the following changes will result in an increase in reaction rate?

(A) decrease size of the graphite pieces
 (B) decrease temperature
 (C) decrease partial pressure of $CO(g)$
 (D) decrease partial pressure of $CO_2(g)$

Kinetics

81. The reaction $\text{CH}_3\text{NC} \rightarrow \text{CH}_3\text{CN}$ is first order. Which reaction characteristic changes as the reaction proceeds?

(A) the half life (B) the rate constant
(C) the rate law (D) the reaction rate



The data below were obtained for the reaction between A and B. What is the rate law for this reaction?

Trial	$[\text{A}]_0$ (mol•L ⁻¹)	$[\text{B}]_0$ (mol•L ⁻¹)	Initial Rate of Reaction (mol•L ⁻¹ •s ⁻¹)
1	0.10	0.10	6.5×10^{-5}
2	0.20	0.10	2.6×10^{-4}
3	0.10	0.20	6.5×10^{-5}

(A) Rate = $k[\text{A}]$ (B) Rate = $k[\text{A}][\text{B}]$
(C) Rate = $k[\text{A}]^2$ (D) Rate = $k[\text{A}]^2[\text{B}]$

83. A first-order reaction is carried out at several different temperatures. The data are plotted as $\ln k$ vs $1/T$. How is the slope of the graph related to E_a ?

(A) $E_a = \text{slope} \times 8.314 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$
(B) $E_a = -\text{slope} \times 8.314 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$
(C) $1/E_a = \text{slope} \times 8.314 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$
(D) $1/E_a = -\text{slope} \times 8.314 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$

84. Which statement about the behavior of a catalyst is correct?

(A) A catalyst reacts with the product and shifts the equilibrium to the right speeding up the reaction.
(B) A catalyst lowers the activation energy of the original reaction pathway.
(C) A catalyst provides additional energy to a reactant so it can achieve the necessary activation energy.
(D) A catalyst provides an alternative reaction pathway with a lower activation energy.

85. For the reaction:

$2\text{A} + 3\text{B} \rightarrow \text{C}$, $[\text{A}]$ is found to decrease at a rate of $2.0 \text{ M}\cdot\text{s}^{-1}$. If the rate law is rate = $k[\text{A}]$, how fast does $[\text{B}]$ decrease under the same conditions?

(A) $0.66 \text{ M}\cdot\text{s}^{-1}$ (B) $1.3 \text{ M}\cdot\text{s}^{-1}$
(C) $2.0 \text{ M}\cdot\text{s}^{-1}$ (D) $3.0 \text{ M}\cdot\text{s}^{-1}$

86. What are the units for the rate constant of a second-order reaction?

(A) s^{-1} (B) $\text{M}^{-1}\cdot\text{s}^{-1}$ (C) $\text{M}^{-2}\cdot\text{s}^{-1}$ (D) $\text{M}^2\cdot\text{s}^{-1}$

87. A sample containing a radioactive isotope produces 2000 counts per minute in a Geiger counter. After 120 hours, the sample produces 250 counts per minute. What is the half-life of the isotope?

(A) 15 h (B) 30 h (C) 40 h (D) 60 h

88. In the rate-limiting approximation for a two-step reaction, the overall rate of the reaction is always equal to the rate of the _____ step in the reaction mechanism.

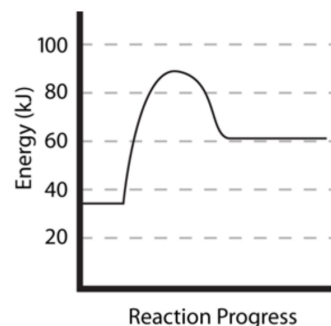
(A) first (B) second
(C) fastest (D) slowest

89. Which of the following examples demonstrate homogeneous catalysis?

I. Pt(s) catalyzing the reaction of $\text{O}_2(\text{g})$ with $\text{CO}(\text{g})$
II. $\text{Cl}(\text{g})$ catalyzing the decomposition of $\text{O}_3(\text{g})$
III. $\text{H}_2\text{O}_2(\text{aq})$ decomposition catalyzed by $\text{Br}^-(\text{aq})$

(A) I. only (B) II. only
(C) I. and III. only (D) II. and III. only

90.



The diagram above represents the energy profile for the reaction: $\text{A} + \text{B} \rightarrow \text{C} + \text{D}$. What is the value of the activation energy for the reaction: $\text{C} + \text{D} \rightarrow \text{A} + \text{B}$?

(A) 25 kJ (B) 55 kJ (C) 85 kJ (D) -30 kJ

91. The rate of a stoichiometric reaction between a solid and a gas in a container may be increased by increasing all of the following factors EXCEPT the

(A) pressure of the gas.
(B) temperature of the gas.
(C) volume of the container.
(D) surface area of the solid.

Kinetics

92. The gas phase decomposition of dinitrogen pentoxide is represented by this equation.



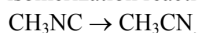
What is the rate of formation of oxygen gas (in $\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$) in an experiment where 0.080 mol of N_2O_5 is consumed in a 4.0 L container every 0.20 seconds?

- (A) 0.020 (B) 0.050 (C) 0.10 (D) 0.20

93. Which accounts for the increase in the rate of a reaction when a catalyst is added to a chemical system?

- (A) a decrease in the enthalpy change between the reactants and products
(B) an increase in the potential energy of the reactants
(C) a decrease in the potential energy of the activated complex
(D) a decrease in the entropy of the activated complex

94. For this first-order isomerization reaction,



how do the properties of the reaction in the table below vary as the reaction proceeds?

	Rate of reaction $-\frac{\Delta[\text{CH}_3\text{NC}]}{\Delta t}$, ($\text{M}\cdot\text{s}^{-1}$)	Half-life, (s)
(A)	remains the same	decreases
(B)	decreases	remains the same
(C)	remains the same	remains the same
(D)	decreases	decreases

- (A) A (B) B (C) C (D) D

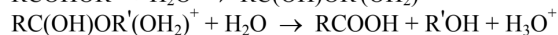
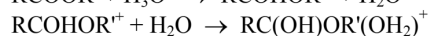
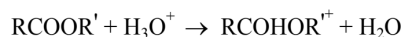
95. The rate of a reaction between A and B follows the rate law $\text{Rate} = k[\text{A}]^2[\text{B}]$.

Determine the rate of Experiment Two in the table below at the same temperature as Experiment One.

Experiment	[A], M	[B], M	Observed Rate, $\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$
1	0.10	0.10	R
2	0.30	0.20	?

- (A) 3R (B) 6R (C) 12R (D) 18R

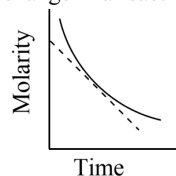
96. One proposed mechanism for the hydrolysis of an ester is shown below.



Which species is considered an intermediate?

- (A) RCOHOR'^+ (B) H_3O^+
(C) RCOOR' (D) R'OH

97. What quantity is represented by the slope of the dashed line in the accompanying diagram for the concentration change in a reaction?

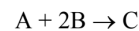


- (A) instantaneous reaction rate
(B) rate constant
(C) reaction order for that reactant
(D) activation energy

98. Changes in which factors affect both the rate and the rate constant of a first order reaction?
I temperature II concentration

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

99. Consider the hypothetical reaction:



The rate remains constant when the concentration of A is doubled and the concentration of B is held constant. The reaction rate doubles when the concentration of B is doubled and the concentration of A is held constant. What are the orders of A and B in this reaction?

- (A) A = 0, B = 1 (B) A = 0, B = 2
(C) A = 1, B = 2 (D) A = 1, B = 0

100. Which statement is true about a reactant that appears in the balanced equation for a reaction but does not appear in the rate equation?

- (A) It is an inhibitor.
(B) It is not part of the reaction.
(C) Its concentration is too low to be important.
(D) It takes part in the reaction after the rate-determining step.

101. Introduction of two drops of concentrated sulfuric acid, H_2SO_4 , speeds up an esterification reaction. Introduction of a piece of platinum metal, Pt, speeds up the reaction of H_2 and O_2 gas. Which of the following statements is true?

- (A) Pt is a homogeneous catalyst; H_2SO_4 is a heterogeneous catalyst.
(B) Pt is a heterogeneous catalyst; H_2SO_4 is a homogeneous catalyst.
(C) Pt and H_2SO_4 are both heterogeneous catalysts.
(D) Pt and H_2SO_4 are both homogeneous catalysts.

Kinetics

102. Which change will decrease the rate of the reaction between $\text{I}_2(\text{s})$ and $\text{H}_2(\text{g})$?
- (A) Increasing the partial pressure of $\text{H}_2(\text{g})$
 (B) Adding the $\text{I}_2(\text{s})$ as one piece rather than as several small ones
 (C) Heating the reaction mixture
 (D) Adding a catalyst for the reaction

103. Factors that can affect the rate of a chemical reaction between a solid and a solution include all of the following EXCEPT the
- (A) concentration of the reactants in solution.
 (B) volume of the container.
 (C) size of the solid particles.
 (D) temperature.

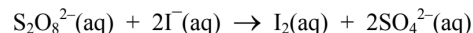
104. The commercial production of ammonia is represented by the equation $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$. If the rate of disappearance of $\text{H}_2(\text{g})$ is $1.2 \times 10^{-3} \text{ mol/min}$, what is the rate of appearance of $\text{NH}_3(\text{g})$?
- (A) $2.4 \times 10^{-3} \text{ mol/min}$ (B) $1.8 \times 10^{-3} \text{ mol/min}$
 (C) $1.2 \times 10^{-3} \text{ mol/min}$ (D) $8.0 \times 10^{-4} \text{ mol/min}$

105. A reaction is endothermic with $\Delta H = 100. \text{ kJ/mol}$. If the activation enthalpy of the forward reaction is $140. \text{ kJ/mol}$, what is the activation enthalpy of the reverse reaction?
- (A) $40. \text{ kJ/mol}$ (B) $100. \text{ kJ/mol}$
 (C) $140. \text{ kJ/mol}$ (D) $240. \text{ kJ/mol}$

106. The first-order disappearance of a substance has a half-life of 34.0 s . How long does it take for the concentration of that substance to fall to 12.5% of its initial value?
- (A) 11 s (B) 68 s (C) 102 s (D) 272 s

107. The reaction $\text{C}_3\text{H}_7\text{I} + \text{Cl}^- \rightarrow \text{C}_3\text{H}_7\text{Cl} + \text{I}^-$ is thought to occur in the polar solvent CH_3OH by the mechanism:
- Step 1 $\text{C}_3\text{H}_7\text{I} \rightarrow \text{C}_3\text{H}_7^+ + \text{I}^-$ (slow)
 Step 2 $\text{C}_3\text{H}_7^+ + \text{Cl}^- \rightarrow \text{C}_3\text{H}_7\text{Cl}$ (fast)
 Which species is an intermediate in this reaction?
 (A) CH_3OH (B) C_3H_7^+ (C) I^- (D) Cl^-

108. The times listed are those recorded at 25°C for the reaction below to produce a measurable amount of $\text{I}_2(\text{aq})$:

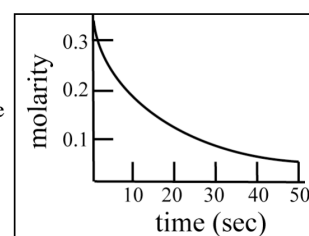


Experiment	Initial $[\text{S}_2\text{O}_8^{2-}]$, M	Initial $[\text{I}^-]$, M	Time (sec)
1	0.0400	0.0800	39
2	0.0400	0.0400	78
3	0.0100	0.0800	156
4	0.0200	0.0200	?

What is the expected time for experiment 4?

- (A) 156 sec (B) 234 sec
 (C) 312 sec (D) 624 sec

109. For the reaction represented by the accompanying diagram, which reaction rate is the greatest?



- (A) average rate (B) final rate
 (C) initial rate (D) rate at 20 seconds

110. Which units are appropriate for a reaction rate?

- (A) $\text{mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$ (B) $\text{mol} \cdot \text{L}^{-1}$
 (C) $\text{mol} \cdot \text{L} \cdot \text{s}^{-1}$ (D) $\text{L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$

111. What is the rate equation for a reaction, $\text{A} + \text{B} \rightarrow \text{products}$, based on the rate data?

$[\text{A}]_0, \text{mol} \cdot \text{L}^{-1}$	$[\text{B}]_0, \text{mol} \cdot \text{L}^{-1}$	Rate
0.15	0.10	x
0.30	0.20	4x
0.30	0.40	16x

- (A) $\text{Rate} = k[\text{A}]^2$ (B) $\text{Rate} = k[\text{B}]^2$
 (C) $\text{Rate} = k[\text{A}][\text{B}]$ (D) $\text{Rate} = k[\text{A}][\text{B}]^2$

112. The effect of temperature on the rates of chemical reactions is primarily a result of the

- (A) size of the colliding molecules.
 (B) orientation of the colliding molecules.
 (C) enthalpies of the reactants and products.
 (D) kinetic energies of the colliding molecules.

Kinetics

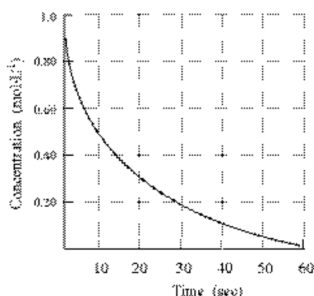
113. The value of the rate constant for a gas phase reaction can be changed by increasing the

(A) amount of product.
(B) pressure of the reactant.
(C) temperature of the reaction vessel.
(D) volume of the reaction vessel.

114. What is the half life of the irreversible first order reaction, $A \rightarrow B$, if 75% of A is converted to B in 60 minutes?

(A) 30 minutes (B) 45 minutes
(C) 60 minutes (D) 80 minutes

115. According to the graph what is the rate of disappearance of the reactant at 10 seconds?



(A) $0.025 \text{ mol}\cdot\text{L}^{-1}\cdot\text{sec}^{-1}$ (B) $0.050 \text{ mol}\cdot\text{L}^{-1}\cdot\text{sec}^{-1}$
(C) $0.40 \text{ mol}\cdot\text{L}^{-1}\cdot\text{sec}^{-1}$ (D) $0.80 \text{ mol}\cdot\text{L}^{-1}\cdot\text{sec}^{-1}$

116. For the reaction, $\text{C}_3\text{H}_8 + 5\text{O}_2 \rightarrow 3\text{CO}_2 + 4\text{H}_2\text{O}$, how does the rate of disappearance of O_2 compare to the rate of production of CO_2 ? The rate of disappearance of O_2 is

(A) the same as the rate of production of CO_2 .
(B) $1/5$ the rate of production of CO_2 .
(C) $3/5$ the rate of production of CO_2 .
(D) $5/3$ the rate of production of CO_2 .

117. When the concentration of a particular reactant is tripled and the initial rate of the reaction increases by a factor of nine, what is the order of the reaction with respect to this reactant?

(A) zero (B) one (C) two (D) three

118. A substance, X, undergoes a first order reaction $X \rightarrow Y$ with a half life of 20 minutes. If the initial concentration of X is 2.0 M, what will its concentration be after 40 minutes?

(A) 0.25 M (B) 0.50 M (C) 1.0 M (D) 1.4 M

119. The chemicals in a lightstick give off light as they react. When the lightstick is placed in warm water the glow increases. This is because the

(A) activation energy for the process is lowered.
(B) average kinetic energy of the reactants increases.
(C) higher temperature catalyzes the reaction.
(D) higher temperature changes the wavelength of light emitted.

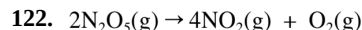
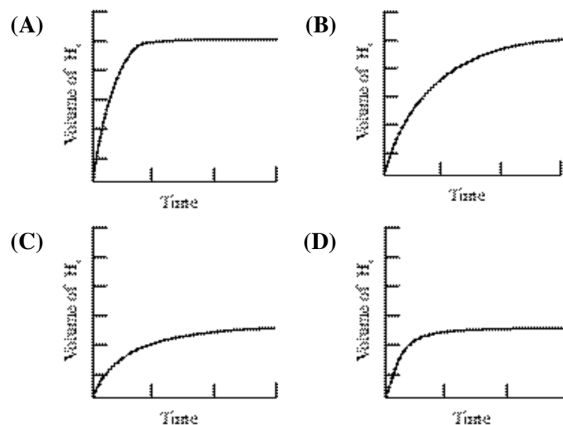
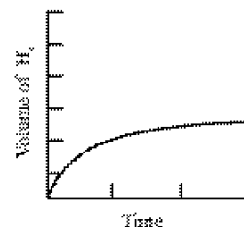
120. Under certain conditions the reaction of CO with NO_2 to give CO_2 and NO results in the rate law;

$$\text{rate} = k[\text{CO}][\text{NO}_2]$$

What are the units for the rate constant, k ?

(A) $\text{mol}\cdot\text{L}^{-1}\cdot\text{min}^{-1}$ (B) $\text{mol}^2\cdot\text{L}^{-2}\cdot\text{min}^{-1}$
(C) $\text{L}\cdot\text{mol}^{-1}\cdot\text{min}^{-1}$ (D) $\text{L}^2\cdot\text{mol}^{-2}\cdot\text{min}^{-1}$

121. The plot shows the volume of H_2 gas produced as a function of time by the reaction of a given mass of magnesium turnings with excess 1 M HCl. What graph results from the reaction of an equal mass of magnesium turnings with excess 2 M HCl? (Assume all graphs are plotted on the same scale as the one shown above.)



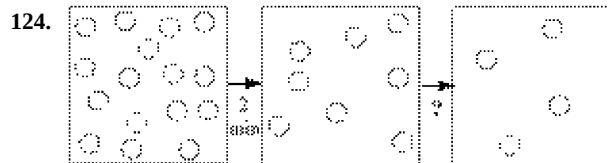
The rate of disappearance of $\text{N}_2\text{O}_5(\text{g})$ at a certain temperature is $0.016 \text{ mol}\cdot\text{L}^{-1}\cdot\text{min}^{-1}$. What is the rate of formation of $\text{NO}_2(\text{g})$ (in $\text{mol}\cdot\text{L}^{-1}\cdot\text{min}^{-1}$) at this temperature?

(A) 0.0080 (B) 0.016 (C) 0.032 (D) 0.064

123. What are the units of the rate constant for a second order reaction?

(A) s^{-1} (B) $\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$
(C) $\text{L}\cdot\text{s}\cdot\text{mol}^{-1}$ (D) $\text{L}\cdot\text{mol}^{-1}\cdot\text{s}^{-1}$

Kinetics



If the disappearance of the spheres in this diagram is a first order process, how much time is required to go from the second box to the third box?

- (A) 1 min (B) 2 min (C) 4 min (D) 8 min

125. A catalyst affects the rate of a chemical reaction by

- (A) increasing the average kinetic energy of the reactants.
(B) increasing the number of collisions between the reactants.
(C) decreasing the energy difference between the reactants and products.
(D) providing an alternate reaction pathway with a lower activation energy.

126. A hypothetical reaction has a rate law of:

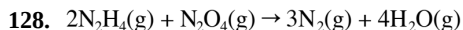
$$\text{Rate} = k[A]^2[B]$$

Which statement about this reaction is most probably correct?

- (A) Doubling the concentration of A will double the rate of the reaction.
(B) Tripling [A] will affect the rate twice as much as tripling [B].
(C) The reaction mechanism involves the formation of B₂ at some stage.
(D) The reaction mechanism involves more than one step.

127. When 100. mL of 1.0 M HCl is added to a 2.0 g piece of CaCO₃, CO₂ is produced at a certain rate. Which of the changes below will NOT increase the rate of this reaction?

- (A) adding 100. mL of 2.0 M HCl in place of 100. mL of 1.0 M HCl
(B) heating the 100. mL of 1.0 M HCl before adding it to the CaCO₃
(C) adding 100. mL of 1.0 M HCl to 2.0 g of powdered CaCO₃
(D) adding 150. mL of 1.0 M HCl in place of 100. mL of 1.0 M HCl



If N₂H₄(g) disappears at a rate of 0.12 mol·L⁻¹·min⁻¹, at what rate does N₂(g) appear?

- (A) 0.080 mol·L⁻¹·min⁻¹ (B) 0.12 mol·L⁻¹·min⁻¹
(C) 0.18 mol·L⁻¹·min⁻¹ (D) 0.36 mol·L⁻¹·min⁻¹

129. Use the data to determine the orders of A and B in the reaction; A + B → products

Experiment	[A], mol·L ⁻¹	[B], mol·L ⁻¹	rate, mol·L ⁻¹ ·s ⁻¹
1	0.10	0.10	0.0090
2	0.20	0.10	0.036
3	0.10	0.20	0.018

- (A) Rate = k[A][B] (B) Rate = k[A]²[B]
(C) Rate = k[A][B]² (D) Rate = k[A]²[B]²



If the activation energy for the forward reaction is 62 kJ, what is the activation energy for the reverse reaction?

- (A) 24 kJ (B) 38 kJ (C) 62 kJ (D) 100 kJ

131. What is the order of a reaction if the rate constant has the units L·mol⁻¹·s⁻¹?

- (A) zero (B) first
(C) second (D) third

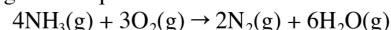
132. An iron catalyst is used in the Haber process in which gaseous N₂ and H₂ react to produce NH₃. What is the role of this catalyst?

- (A) It provides a pathway with a lower activation energy.
(B) It increases the equilibrium constant of the reaction.
(C) It raises the kinetic energies of the reactants.
(D) It interacts with the NH₃.

133. All of the following are expected to affect the rate of an irreversible chemical reaction EXCEPT

- (A) adding a catalyst.
(B) removing some products.
(C) increasing the temperature.
(D) decreasing the reactant concentration.

134. The oxidation of ammonia produces nitrogen and water according to the equation:



If the rate of formation of N₂ at a certain temperature is 3.0 mol·L⁻¹·s⁻¹, what is the rate of disappearance of O₂?

- (A) 2.0 mol·L⁻¹·s⁻¹ (B) 3.0 mol·L⁻¹·s⁻¹
(C) 4.5 mol·L⁻¹·s⁻¹ (D) 9.0 mol·L⁻¹·s⁻¹

135. What are the units of the rate constant for a second order reaction when the rate is expressed in mol·L⁻¹·s⁻¹?

- (A) s⁻¹ (B) L·mol⁻¹
(C) L·mol⁻¹·s⁻¹ (D) L²·mol⁻²·s⁻¹

Kinetics

136. For the reaction:

$(\text{CH}_3)_3\text{CBr}(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow (\text{CH}_3)_3\text{COH}(\text{aq}) + \text{Br}^-(\text{aq})$
it is found that halving the concentration of $(\text{CH}_3)_3\text{CBr}$ causes the reaction rate to be halved but halving the concentration of OH^- has no effect on the rate. What is the rate law?

- (A) $\text{Rate} = k[(\text{CH}_3)_3\text{CBr}]^{1/2}[\text{OH}^-]$
(B) $\text{Rate} = k[(\text{CH}_3)_3\text{CBr}]^2[\text{OH}^-]$
(C) $\text{Rate} = k[(\text{CH}_3)_3\text{CBr}]^{1/2}$
(D) $\text{Rate} = k[(\text{CH}_3)_3\text{CBr}]$

137. A catalyst speeds up a chemical reaction by

- (A) shifting the equilibrium.
(B) increasing the activation energy.
(C) decreasing the reaction enthalpy.
(D) providing an alternate reaction pathway.

138. The activation energy for a reaction can be determined by measuring the reaction rate at different

- (A) temperatures.
(B) catalyst concentrations.
(C) reactant concentrations.
(D) times on the reaction curve.

139. For a rate law of the form; $\text{Rate} = k[\text{A}]^m[\text{B}]^n$, the exponents m and n are obtained from

- (A) changes in rate with changing temperature.
(B) the coefficients of A and B in the balanced equation.
(C) the concentrations of A and B in a single experiment.
(D) changes in the reaction rate for different concentrations of A and B.

140. What is the order of a reaction for which the units of k are $\text{L} \cdot \text{mol}^{-1} \cdot \text{s}^{-1}$ and the units of the rate are $\text{mol} \cdot \text{L}^{-1} \cdot \text{s}^{-1}$?

- (A) zero order (B) first order
(C) second order (D) some other order

141. For the reaction $\text{A} + \text{B} \rightarrow \text{C}$, the rate law is: $\text{Rate} = k[\text{A}]^2$.

Which change(s) will increase the rate of the reaction?

- | | |
|----|-----------------------------------|
| I | Increasing the concentration of A |
| II | Increasing the concentration of B |

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

142. Which does NOT change with time for a first-order reaction?

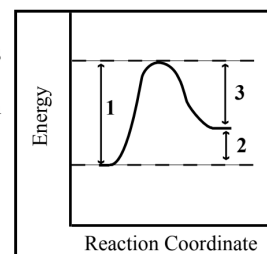
- (A) the amount of reactant that disappears in each half-life
(B) the concentration of the reactant
(C) the length of each half-life
(D) the rate of the reaction

143. The rates of which reactions are increased when the temperature is raised?

- | | |
|----|-----------------------|
| I | endothermic reactions |
| II | exothermic reactions |

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

144. When a catalyst is added to the system represented by this energy-reaction coordinate diagram, which dimensions in the diagram are changed?



- (A) 1 and 2 only (B) 1 and 3 only
(C) 2 and 3 only (D) 1, 2, 3

145. A homogeneous liquid reaction mixture is often heated to increase the rate of reaction. This is best explained by the fact that raising the temperature

- (A) increases the heat of reaction.
(B) decreases the energy of activation.
(C) increases the vapor pressure of the liquid
(D) increases the average kinetic energy of the reactants.

146. For the reaction,
 $2\text{A} + \text{B} \rightarrow \text{C}$
which relationship is correct?

- (A) $\Delta[\text{A}] = \Delta[\text{C}]$ (B) $-\Delta[\text{A}] = \Delta[\text{C}]$
(C) $-2\Delta[\text{A}] = \Delta[\text{C}]$ (D) $-\Delta[\text{A}] = 2\Delta[\text{C}]$

147. This exothermic reaction is catalyzed by $\text{MnO}_2(\text{s})$.



Which of the following will increase the rate of this reaction?

- | | |
|----|---|
| 1. | Raising the temperature |
| 2. | Increasing the surface area of $\text{MnO}_2(\text{s})$ |

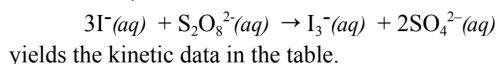
- (A) 1 only (B) 2 only
(C) Both 1 and 2 (D) Neither 1 and 2

Kinetics

148. Which is constant for different reactant concentrations in a first-order reaction?

- (A) The time required for the concentration of reactants to drop below 0.001 M.
 (B) The time required for one-half of reactants to disappear.
 (C) The rate of disappearance of reactants in $\text{mol}\cdot\text{L}^{-1}\cdot\text{time}^{-1}$.
 (D) The rate of formation of products in $\text{mol}\cdot\text{L}^{-1}\cdot\text{time}^{-1}$.

149. The reaction,



$[\text{I}^{-}]_0$ ($\text{mol}\cdot\text{L}^{-1}$)	$[\text{S}_2\text{O}_8^{2-}]_0$ ($\text{mol}\cdot\text{L}^{-1}$)	Relative Rate
0.001	0.001	1
0.002	0.001	2
0.002	0.002	4

What is the rate equation?

- (A) $\text{Rate} = k[\text{I}^{-}][\text{S}_2\text{O}_8^{2-}]$ (B) $\text{Rate} = k[\text{I}^{-}]^2[\text{S}_2\text{O}_8^{2-}]$
 (C) $\text{Rate} = k[\text{I}^{-}]^3[\text{S}_2\text{O}_8^{2-}]$ (D) $\text{Rate} = k[\text{I}^{-}]^2[\text{S}_2\text{O}_8^{2-}]^2$

150. The radioisotope, N-13, has a half-life of 10.0 minutes. What is the rate constant for the radioactive decay of N-13?

- (A) 0.0301 min^{-1} (B) 0.0693 min^{-1}
 (C) 0.100 min^{-1} (D) 6.93 min^{-1}

For the reaction $2\text{C}_2\text{H}_6(\text{g}) + 7\text{O}_2(\text{g}) \rightarrow 4\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{l})$, the rate of disappearance of $\text{C}_2\text{H}_6(\text{g})$

- (A) equals the rate of disappearance of $\text{O}_2(\text{g})$.
 (B) is seven times the rate of disappearance of $\text{O}_2(\text{g})$.
 (C) is twice the rate of appearance of $\text{CO}_2(\text{g})$.
 (D) is one-third the rate of appearance of $\text{H}_2\text{O}(\text{l})$.

152. The rate law for a certain reaction is found to be:

$$\text{rate} = k[\text{A}][\text{B}]^2$$

How will the rate of this reaction compare if the concentration of A is doubled and the concentration of B is halved? The rate will

- (A) remain the same.
 (B) be double the original rate.
 (C) be triple the original rate.
 (D) be one-half the original rate.

153. Use the experimental data in this table to determine the rate law for the reaction of hydrogen iodide, HI, with ethyl iodide, $\text{C}_2\text{H}_5\text{I}$.

$[\text{HI}]$, M	$[\text{C}_2\text{H}_5\text{I}]$, M	Rate, $\text{M}\cdot\text{s}^{-1}$
0.010	0.010	1.2×10^{-5}
0.010	0.020	2.4×10^{-5}
0.020	0.030	7.2×10^{-5}

- (A) $\text{rate} = k[\text{HI}]$
 (B) $\text{rate} = k[\text{HI}][\text{C}_2\text{H}_5\text{I}]$
 (C) $\text{rate} = k[\text{HI}]^2[\text{C}_2\text{H}_5\text{I}]$
 (D) $\text{rate} = k[\text{HI}]^2[\text{C}_2\text{H}_5\text{I}]^3$

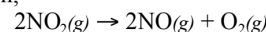
For the reaction $\text{NO}_2(\text{g}) + \text{CO}(\text{g}) \rightarrow \text{NO}(\text{g}) + \text{CO}_2(\text{g})$ at temperatures below 500 K, the rate law is $\text{rate} = k[\text{NO}_2]^2$. Which mechanism is consistent with this information?

Mechanism 1 $\text{NO}_2 + \text{NO}_2 \rightarrow \text{NO}_3 + \text{NO}$ slow
 $\text{CO} + \text{NO}_3 \rightarrow \text{CO}_2 + \text{NO}_2$ fast

Mechanism 2 $\text{NO}_2 + \text{NO}_2 \rightleftharpoons \text{NO}_3 + \text{NO}$ fast
 $\text{CO} + \text{NO}_3 \rightarrow \text{CO}_2 + \text{NO}_2$ slow

- (A) 1 only (B) 2 only
 (C) either 1 or 2 (D) neither 1 nor 2

155. For the reaction;



at a certain temperature, the initial rate of decomposition of NO_2 is $0.0036 \text{ mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$. What is the initial rate of formation of $\text{O}_2(\text{g})$ in $\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$?

- (A) 0.0018 (B) 0.0036
 (C) 0.0054 (D) 0.0072

156. The reaction $\text{A} + \text{B} \rightarrow \text{AB}$ has an enthalpy of reaction of $-85.0 \text{ kJ}\cdot\text{mol}^{-1}$. If the activation enthalpy for the forward reaction is $120.0 \text{ kJ}\cdot\text{mol}^{-1}$, what is the activation energy for the reverse reaction $\text{AB} \rightarrow \text{A} + \text{B}$?

- (A) $35.0 \text{ kJ}\cdot\text{mol}^{-1}$ (B) $85.0 \text{ kJ}\cdot\text{mol}^{-1}$
 (C) $120.0 \text{ kJ}\cdot\text{mol}^{-1}$ (D) $205.0 \text{ kJ}\cdot\text{mol}^{-1}$

157. Use the experimental data in this table to determine the rate law for the reaction of hydrogen iodide, HI, with ethyl iodide, $\text{C}_2\text{H}_5\text{I}$, to form ethane, C_2H_6 , and iodine, I_2 .

$[\text{HI}]$, M	$[\text{C}_2\text{H}_5\text{I}]$, M	$\frac{\Delta[\text{I}_2]}{\Delta t}$, $\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$
0.010	0.010	1.2×10^{-5}
0.010	0.020	2.4×10^{-5}
0.020	0.020	4.8×10^{-5}

- (A) $\text{rate} = k[\text{HI}]$ (B) $\text{rate} = k[\text{C}_2\text{H}_5\text{I}]$
 (C) $\text{rate} = k[\text{HI}][\text{C}_2\text{H}_5\text{I}]$ (D) $\text{rate} = k[\text{HI}]^2[\text{C}_2\text{H}_5\text{I}]^2$

Kinetics

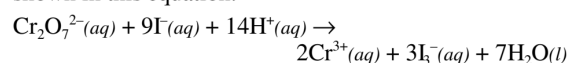
158. Which equation indicates the presence of a catalyst in the reaction?

(A) $A + B \rightarrow D + B$ (B) $A + B \rightarrow C + D$
 (C) $A + A \rightarrow D$ (D) $A \rightarrow B + C$

159. The reaction $A \rightarrow B + C$ obeys the rate law: rate = $k[A]$. Which graph will produce a straight line?

(A) $[A]$ vs time (B) $\ln[A]$ vs time
 (C) $1/[A]$ vs time (D) $[A]$ vs $1/\text{time}$

160. Iodide ion is oxidized by acidified dichromate ions as shown in this equation.



These data were obtained when the reaction was studied at a constant pH.

Experiment	$[\text{Cr}_2\text{O}_7^{2-}]$, M	$[\text{I}^-]$, M	Rate, $\text{M}\cdot\text{s}^{-1}$
1	0.0040	0.010	0.00050
2	0.0080	0.010	0.0010
3	0.0120	0.020	0.0060

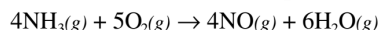
What is the order of the reaction with respect to $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$ and $\text{I}^-(\text{aq})$?

- (A) first order with respect to both $\text{Cr}_2\text{O}_7^{2-}$ and I^-
 (B) second order with respect to both $\text{Cr}_2\text{O}_7^{2-}$ and I^-
 (C) second order with respect to $\text{Cr}_2\text{O}_7^{2-}$ and first order with respect to I^-
 (D) first order with respect to $\text{Cr}_2\text{O}_7^{2-}$ and second order with respect to I^-

161. The reaction $A \rightarrow B$ is first order in A. Which plot will be linear?

(A) $[A]$ vs. time (B) $\ln [A]$ vs. time
 (C) $1/[A]^2$ vs. time (D) $1/[A]$ vs. time

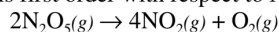
162. One of the steps in the manufacture of nitric acid is the oxidation of ammonia shown in this equation.



If gaseous water appears at a rate of $0.025 \text{ mol}\cdot\text{min}^{-1}$, at what rate does ammonia disappear?

- (A) $0.0040 \text{ mol}\cdot\text{min}^{-1}$ (B) $0.017 \text{ mol}\cdot\text{min}^{-1}$
 (C) $0.038 \text{ mol}\cdot\text{min}^{-1}$ (D) $0.150 \text{ mol}\cdot\text{min}^{-1}$

163. This reaction is first order with respect to N_2O_5 .



If the half-life for this reaction is 19.0 minutes, what is the rate constant, k ?

- (A) 0.0158 min^{-1} (B) 0.0263 min^{-1}
 (C) 0.0365 min^{-1} (D) 0.0526 min^{-1}